

INFORMATION TECHNOLOGY INSTITUTE

Prelab

Parallel and Series RLC Resonance

AC analysis, resonance verification, Q, bandwidth, sweeps, and transient response





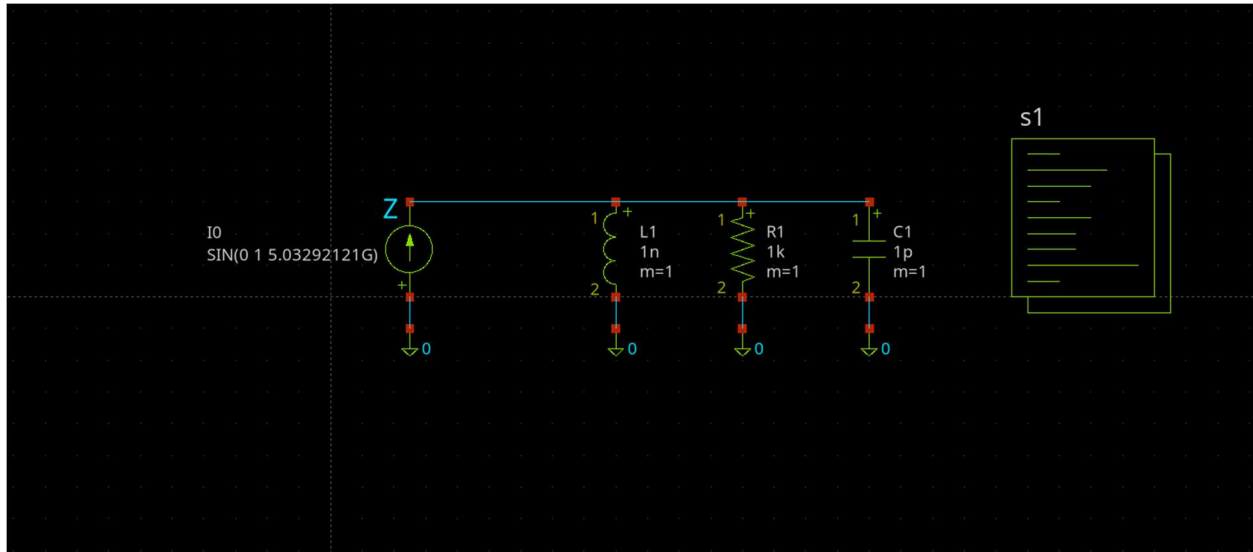
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Part 1 parallel RLC circuit

Schematic



A) Zin Expression

$$Y_{IN}(s) = \frac{1}{R} + \frac{1}{sL} + sC \quad \text{so} \quad Z_{IN}(s) = \left(\frac{1}{R} + \frac{1}{sL} + sC \right)^{-1} = \frac{sLR}{s^2LCR + sL + R}$$

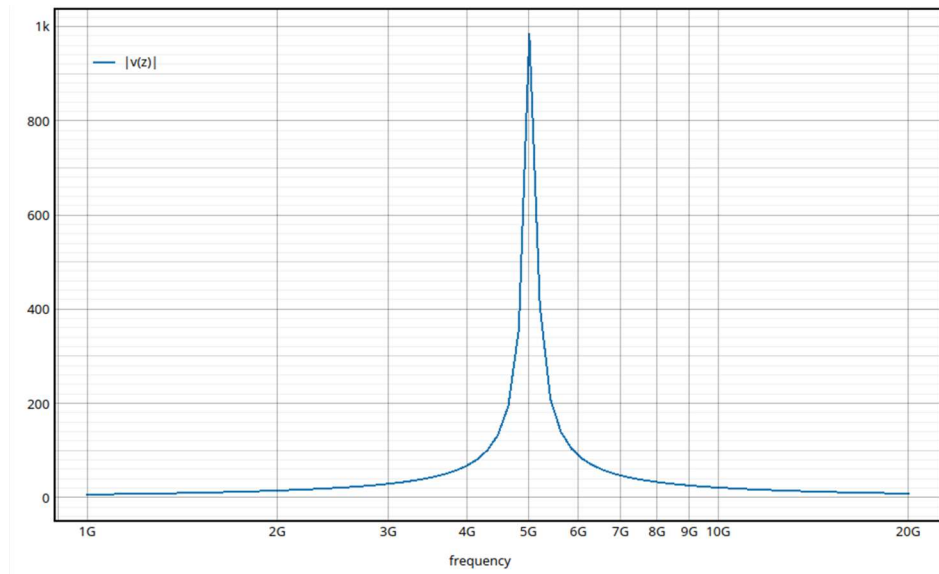
$$Z_{IN}(j\omega) = \frac{1}{\frac{1}{R} + j\left(\omega C - \frac{1}{\omega L}\right)}, \quad Z_{IN}(f) = \frac{1}{\frac{1}{R} + j\left(2\pi fC - \frac{1}{2\pi fL}\right)}$$

$$|Z_{IN}(f)| = \frac{1}{\sqrt{\left(\frac{1}{R}\right)^2 + \left(2\pi fC - \frac{1}{2\pi fL}\right)^2}}$$

$$f_0 = \frac{1}{2\pi\sqrt{LC}}, \quad Z_{IN}(f_0) = R$$

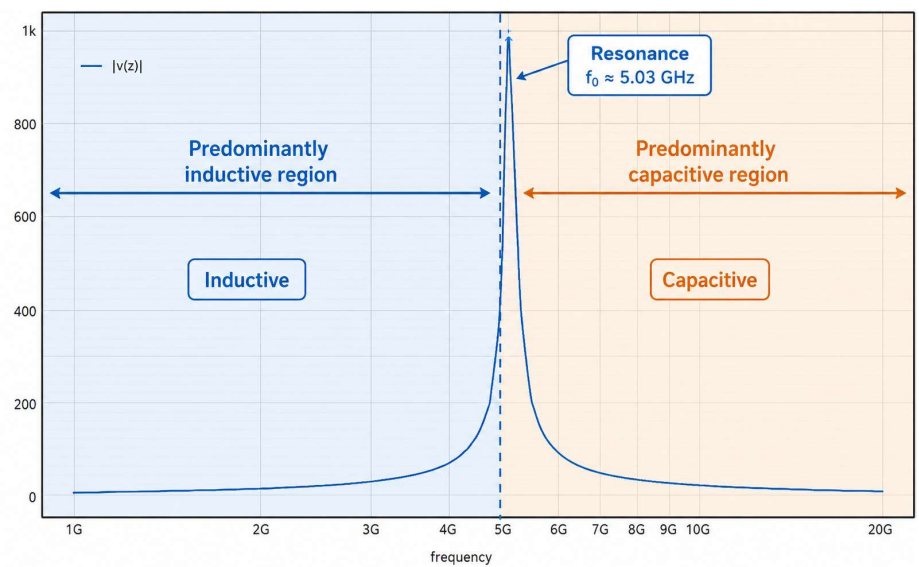


B) AC Analysis ($|Z_{IN}|$ vs. Frequency)



Before the f_0 it is Inductive after f_0

It is capacitive





C) Resonance Frequency Verification

Analytical:

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

$$f_0 = 5.03292 \text{ GHz}$$

Simulation

```
No. of Data Rows : 79
z_peak          = 9.834383e+02 at= 5.018277e+09
f_peak          = 5.018277e+09 with= 9.834383e+02
Peak impedance:
z_peak = 9.834383e+02
Resonance frequency:
f_peak = 5.018277e+09
binary raw file "resonance.raw"
binary raw file "/home/ubuntu/.xschem/simulations/prelab.raw"
Doing analysis at TEMP = 27.000000 and TNOM = 27.000000
```

D) Quality Factor Evaluation

Analytical

$$Q = \frac{R}{W_0 L} = 31.6228$$

simulation

$$Q = 31.60388$$

```
No. of Data Rows : 2603
z_peak          = 9.995927e+02 at= 5.035194e+09
f_peak          = 5.035194e+09 with= 9.995927e+02
f_low           = 4.953897e+09
f_high          = 5.113219e+09
Quality-factor results:
z_peak = 9.9959270000e+02
z_3db = 7.0681877659e+02
f_peak = 5.0351940000e+09
f_low = 4.9538970000e+09
f_high = 5.1132190000e+09
bandwidth = 1.5932200000e+08
q_analytical = 3.1622776602e+01
q_sim = 3.1603883958e+01
q_error_pct = 5.9743784199e-02
binary raw file "quality_factor.raw"
binary raw file "/home/ubuntu/.xschem/simulations/prelab.raw"
Doing analysis at TEMP = 27.000000 and TNOM = 27.000000
```

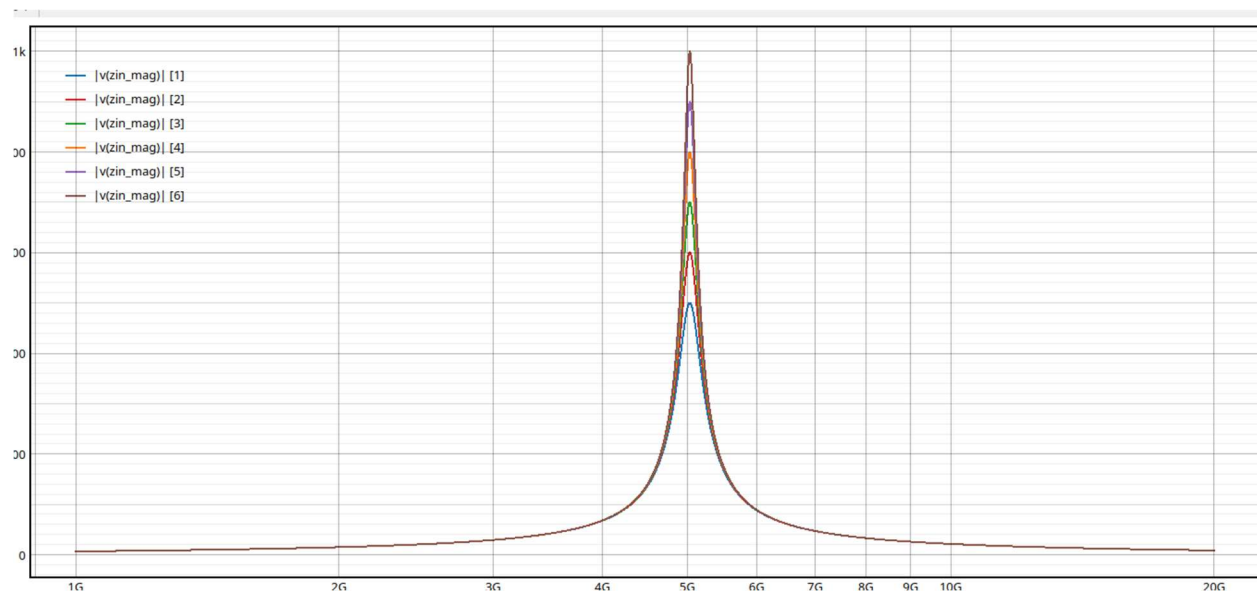


E) Bandwidth (BW)

```
z_peak      = 9.995927e+02 at= 5.035194e+09
f_peak      = 5.035194e+09 with= 9.995927e+02
f_low       = 4.953897e+09
f_high      = 5.113219e+09
Quality-factor results:
z_peak = 9.9959270000e+02
z_3db = 7.0681877659e+02
f_peak = 5.0351940000e+09
f_low = 4.9538970000e+09
f_high = 5.1132190000e+09
bandwidth = 1.5932200000e+08
q_analytical = 3.1622776602e+01
q_sim = 3.1603883958e+01
q_error_pct = 5.9743784199e-02
```

BW=159.322MHz

F) Parametric Sweep on R



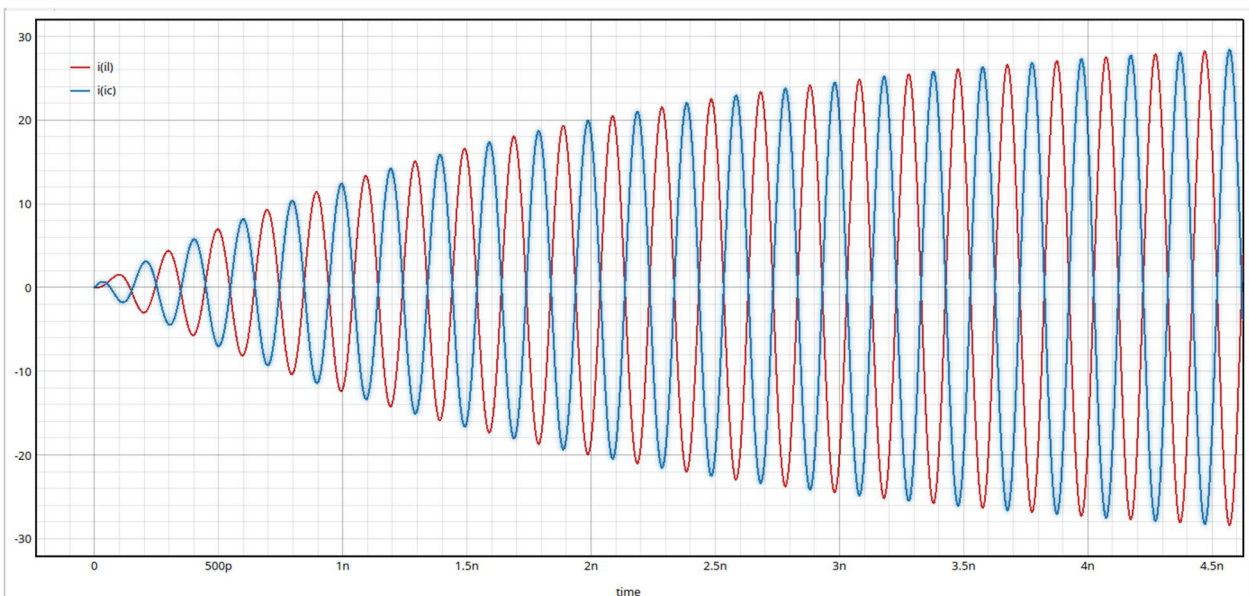


R (Ω)	Peak Z _{IN} (Ω)	f _{peak} (GHz)	f _L (GHz)	f _H (GHz)	Bandwidth (MHz)	Q Analytical	Q Simulation	Error (%)
500	499.949	5.035194	4.876245	5.194630	318.385	15.8114	15.8148	0.0216
600	599.912	5.035194	4.901989	5.167354	265.365	18.9737	18.9746	0.00491
700	699.860	5.035194	4.920477	5.147948	227.471	22.1359	22.1355	0.00181
800	799.791	5.035194	4.934362	5.133444	199.082	25.2982	25.2921	0.0244
900	899.703	5.035194	4.945201	5.122187	176.986	28.4605	28.4497	0.0380
1000	999.593	5.035194	4.953897	5.113219	159.322	31.6228	31.6039	0.0597

comment :

The parametric sweep shows that the resonance frequency remains almost constant at approximately 5.035GHz, since it is determined mainly by L and C, which are unchanged. As the parallel resistance increases from 500 Ω to 1k Ω , the peak input impedance increases from approximately 500 Ω to 1k Ω . The quality factor increases from approximately 15.81 to 31.62, while the bandwidth decreases from approximately 318.4MHz to 159.3MHz. Consequently, increasing R makes the resonance peak taller, narrower, and sharper. The close agreement between the analytical and simulated quality factors, with errors below 0.06%, verifies the theoretical relation $Q=R*\sqrt{C/L}$.

G) Transient response at Resonance





```
vz_max      = 9.992110e+02 at= 1.495236e-08
vz_min      = -9.991150e+02 at= 1.465436e-08
ic_max      = 3.160759e+01 at= 1.490236e-08
ic_min      = -3.160487e+01 at= 1.460436e-08
il_max      = 3.158393e+01 at= 1.460436e-08
il_min      = -3.158664e+01 at= 1.490236e-08

Steady-state results:
vz_peak = 9.9916300000e+02
vz_pp = 1.9983260000e+03
vz_rms = 7.0651493281e+02
ic_peak = 3.1606230000e+01
il_peak = 3.1585285000e+01
```

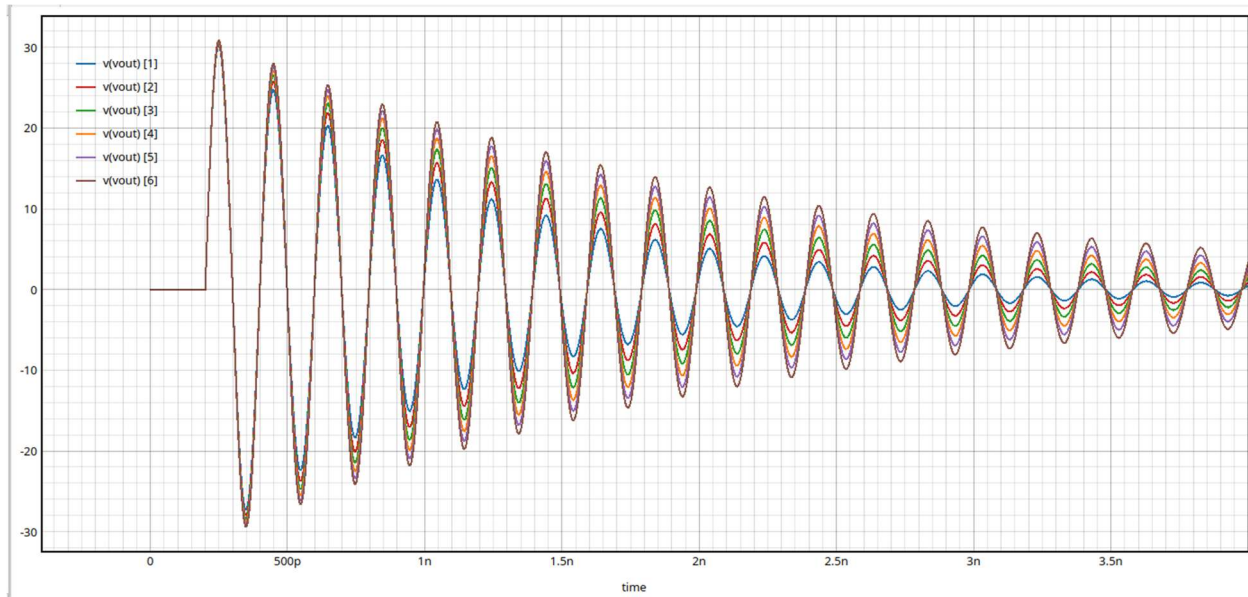
comment :

At resonance, the capacitor and inductor currents are almost equal, about 31.6 A , and 180° out of phase. This means they mostly cancel each other at the input.

The steady-state output voltage is about 999 V , which is very close to the theoretical value of 1000 V .

The large currents do not mean the passive circuit creates power. They are caused by energy moving back and forth between the capacitor and inductor.

H) Step Response:





Comment :

As R increases, the circuit becomes less damped, so the oscillations last longer and the settling time increases. The $500\ \Omega$ case settles the fastest, while the $1\ k\Omega$ case shows the strongest and longest ringing.

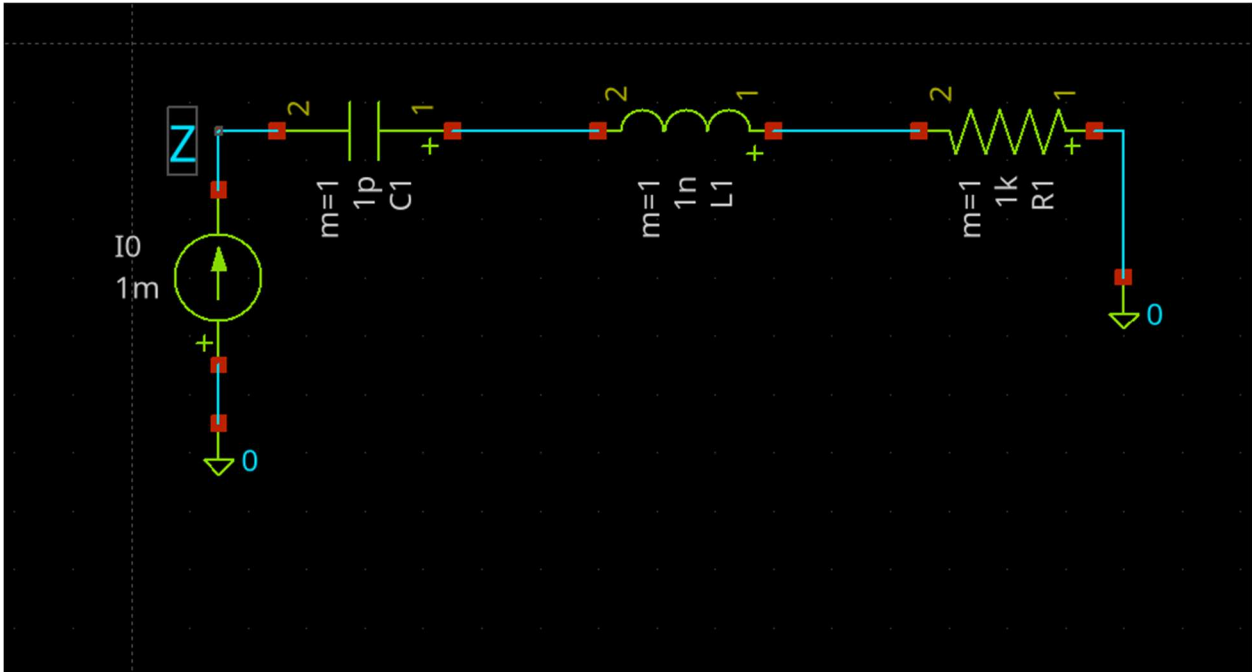
All tested values produce an underdamped response, and the oscillation frequency stays almost the same because L and C are unchanged.

Resistance, (R) (Ω)	Maximum output voltage	Minimum output voltage	Peak-to-peak voltage
500	30.11858	-27.26729	57.38587
600	30.35964	-27.94427	58.30391
700	30.53350	-28.44080	58.97430
800	30.66482	-28.81924	59.48406
900	30.76751	-29.11720	59.88471
1000	30.85001	-29.35789	60.20790



Part 2 Series RLC circuit

Schematic



A) Input Impedance Expression

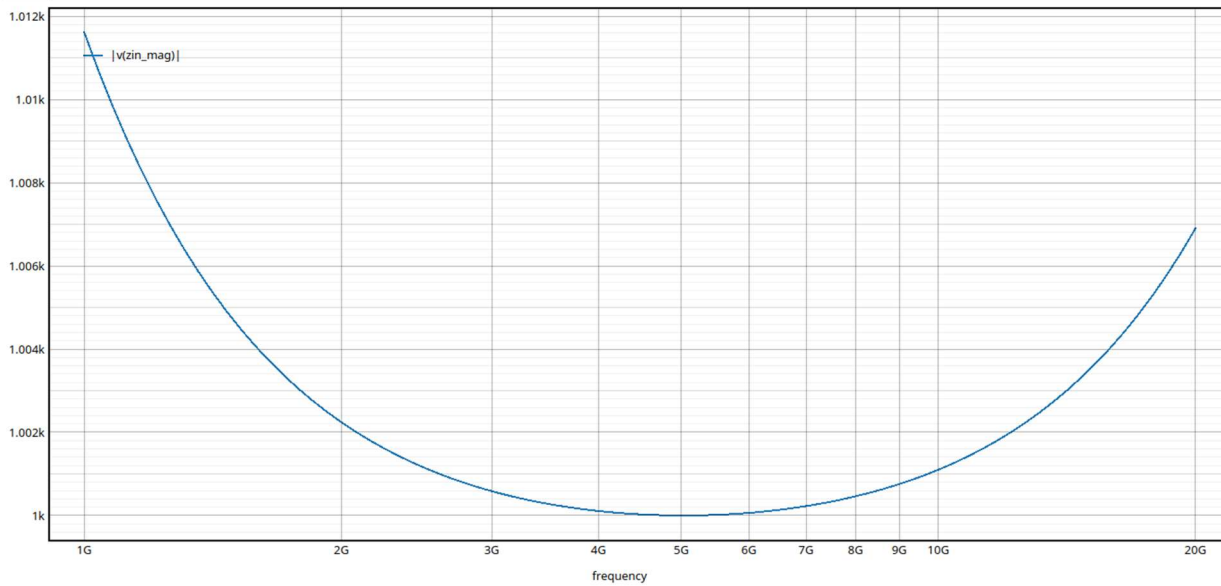
$$\begin{aligned} Z_{IN}(s) &= R + sL + \frac{1}{sC} \\ &= \frac{LCs^2 + RCs + 1}{sC} \\ Z_{IN}(j\omega) &= R + j\left(\omega L - \frac{1}{\omega C}\right) \end{aligned}$$

$$|Z_{IN}| = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

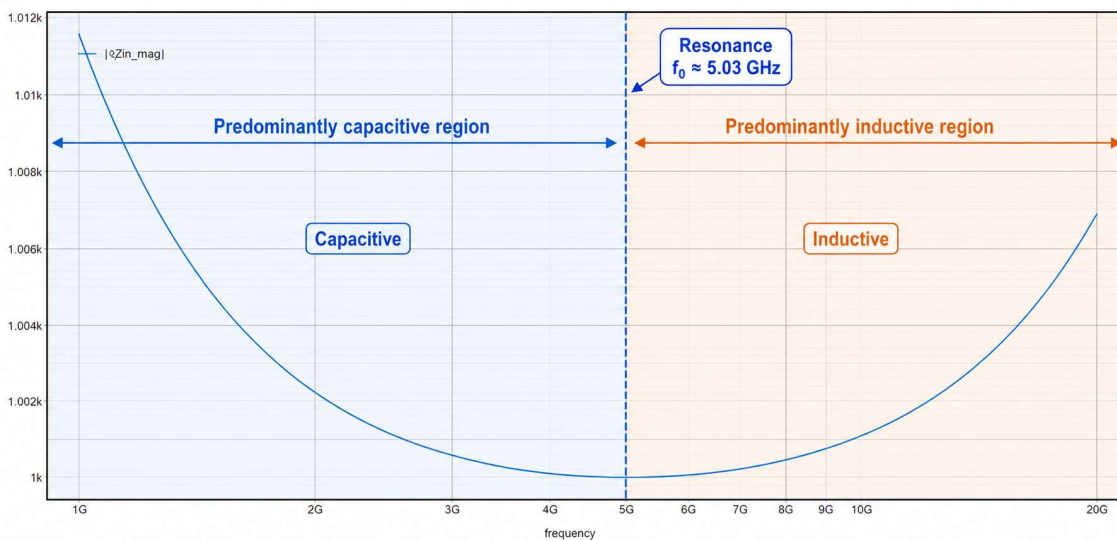
$$f_0 = \frac{1}{2\pi\sqrt{LC}} = 5.0329 \text{ GHz}$$



B) AC Analysis (/ZIN / vs. Frequency)



No. of Data Rows : 79
z_min = 1.000000e+03 at= 5.018277e+09
f_res = 5.018277e+09 with= 1.000000e+03
Minimum input impedance:
z_min = 1.0000000000e+03
Simulated resonance frequency:
f_res = 5.0182770000e+09





C) Resonance Frequency Verification

Analytical

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = 5.0329 \text{ GHz}$$

Simulation

```
No. of Data Rows : 79
z_min           = 1.000000e+03 at= 5.018277e+09
f_res           = 5.018277e+09 with= 1.000000e+03

Minimum input impedance:
z_min = 1.0000000000e+03
Simulated resonance frequency:
f_res = 5.0182770000e+09
```

$$f_0 = 5.018277 \text{ GHz}$$

D) Quality Factor Evaluation

Simulation

```
z_min           = 1.000000e+03 at= 5.035006e+09
f_res           = 5.035006e+09 with= 1.000000e+03
f_low           = 1.589972e+08
f_high          = 1.593131e+11
Extracted quality factor:
q_sim = 3.1636042750e-02
```

Analytical

$$Q = \frac{1}{R} \sqrt{\frac{L}{C}}$$

$$Q = 0.03162$$



E) Bandwidth (BW)

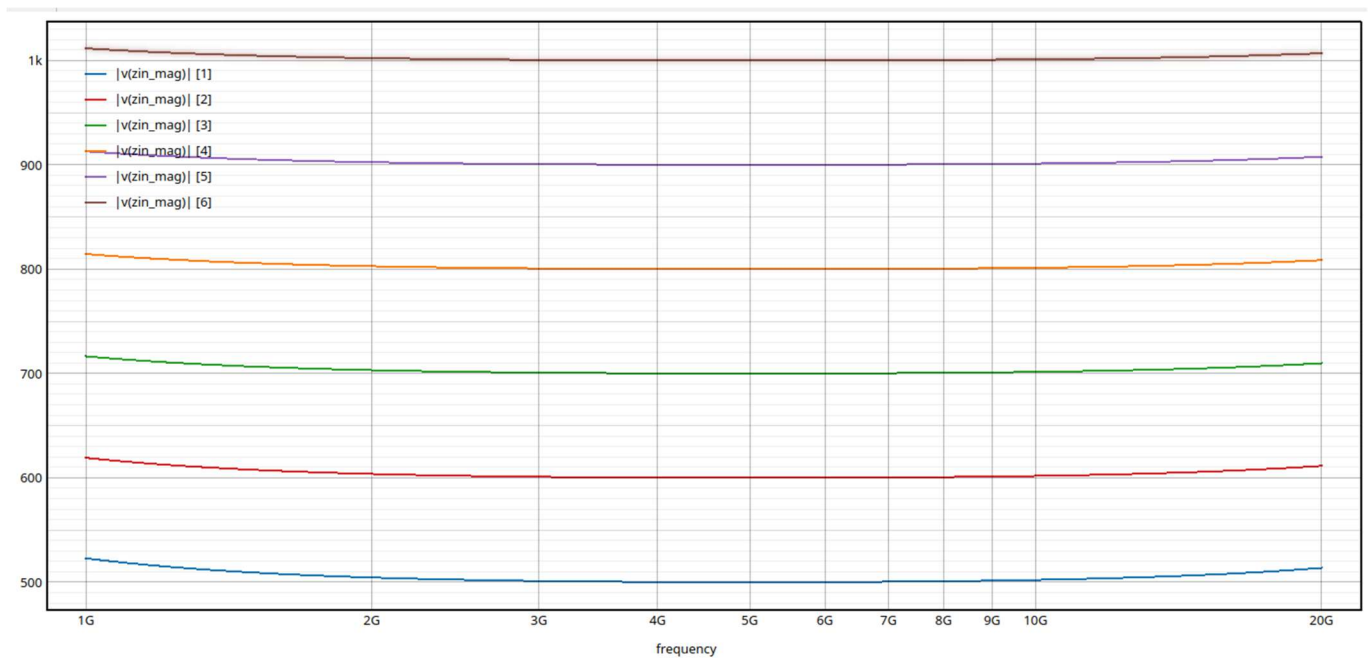
```
z_min      = 1.000000e+03 at= 5.035006e+09  
f_low      = 1.589972e+08  
f_high     = 1.593131e+11  
f_low = 1.5899720000e+08  
f_high = 1.5931310000e+11  
bandwidth = 1.5915410280e+11
```

BW=159.154GHz

$$BW = \frac{f_0}{Q}$$

$$Q = \frac{1}{R} \sqrt{\frac{L}{C}} \quad BW = \frac{R}{2\pi L} = 159.15 \text{ GHz}$$

F) Parametric Sweep on R





Resistance (R)	Quality Factor (Q)	Resonance Dip
500	0.06325	Sharpest and deepest
600	0.05270	Slightly less sharp
700	0.04518	Broader
800	0.03953	Less pronounced
900	0.03514	Very broad
1000	0.03162	Broadest and flattest

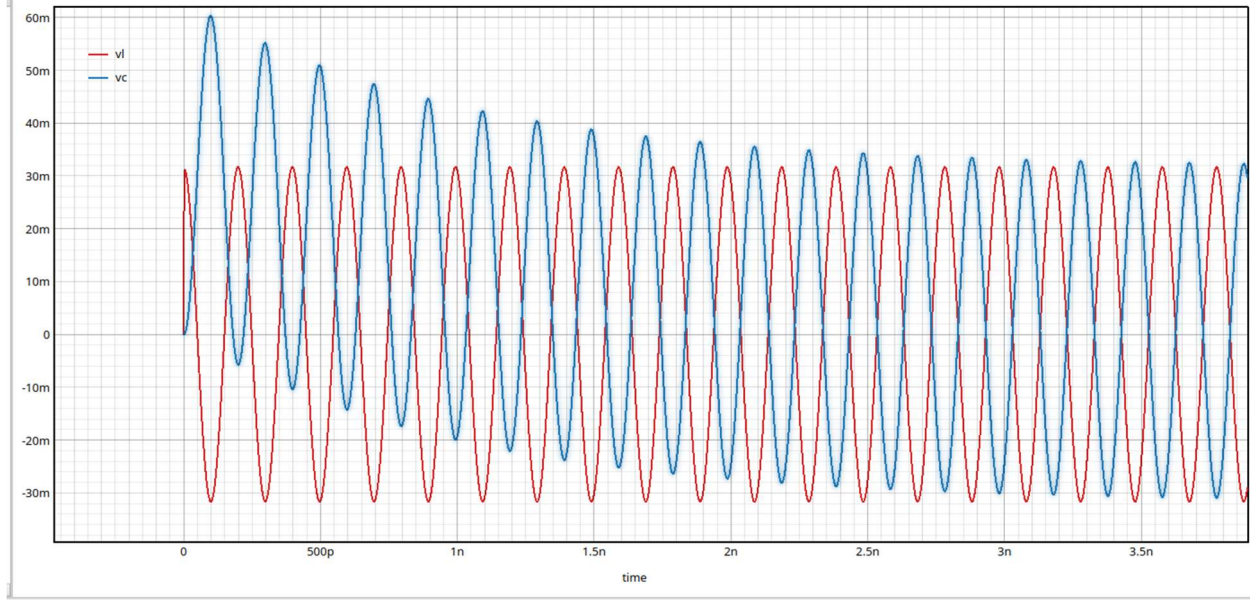
Comment:

As R increases from 500 Ω to 1 k Ω , the quality factor decreases because

$$Q = \frac{1}{R} \sqrt{\frac{L}{C}}$$

Therefore, the resonance dip becomes broader and less sharp. The 500 Ω case has the highest Q and the sharpest dip, while the 1 k Ω case has the lowest Q and the flattest dip. The resonance frequency stays almost unchanged because it depends mainly on L and C .

G) Transient response at Resonance



```
vc_max      = 3.162984e-02 at= 8.047166e-09
vc_min      = -3.161786e-02 at= 9.537166e-09
vL_max      = 3.162540e-02 at= 8.941166e-09
vL_min      = -3.162541e-02 at= 9.239166e-09
Capacitor voltage results:
vc_peak = 3.1623850000e-02
vc_pp = 6.3247700000e-02
vc_rms = 2.2361438782e-02
Inductor voltage amplitude:
vL_peak = 3.1625405000e-02
```

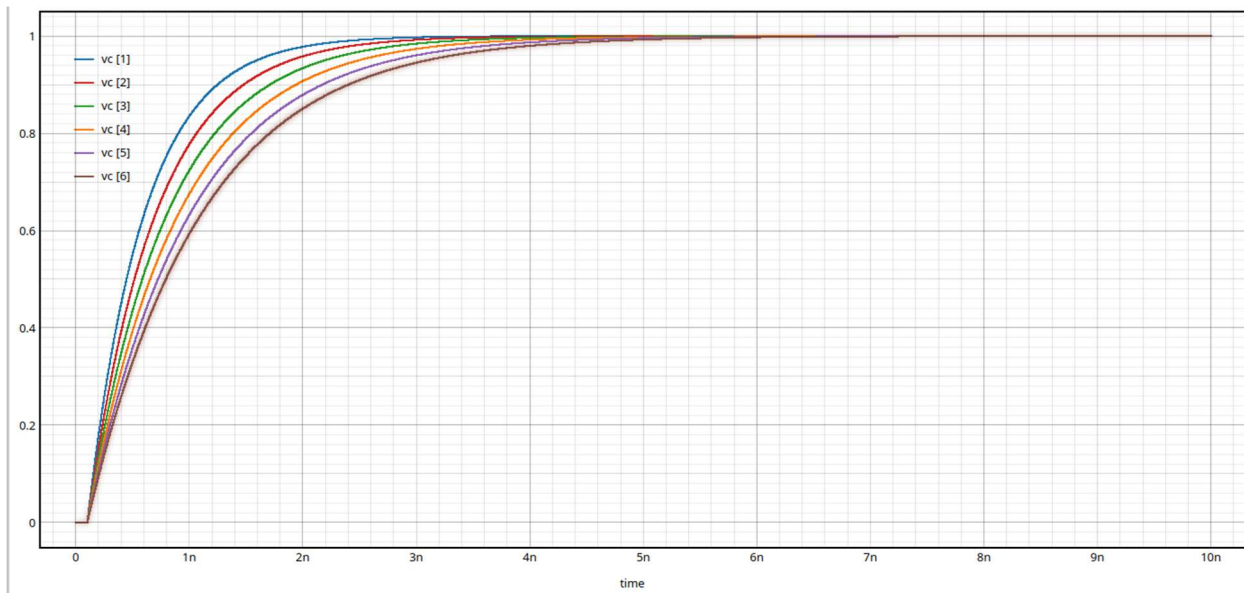
Comment:

At the beginning, the capacitor voltage is temporarily larger because the circuit is still building up energy. This startup effect gradually dies out.

Once steady state is reached, V_C and V_L have almost the same amplitude and are 180° out of phase, so they cancel each other. The resistor voltage then becomes approximately equal to the 1 Vsource voltage. In this circuit, the capacitor and inductor voltages remain below the source voltage because the quality factor is low.



H) Step Response



Resistance (R)	Damping Ratio	Damping Behavior
500	7.9057	Overdamped
600	9.4868	Overdamped
700	11.0680	Overdamped
800	12.6491	Overdamped
900	14.2302	Overdamped
1000	15.8114	Overdamped

Comment:

All resistor values give an overdamped response because $\zeta > 1$. Therefore, the output reaches its final value without oscillation or overshoot. As R increases, the damping becomes stronger and the response becomes slower. The 500 Ω case responds the fastest, while the 1 k Ω case takes the longest time to settle.